

Software Test Plan

Hostel Management System (HMS)

Team 1

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1. Introduction

Purpose: This document defines the test plan for HMS v1.0. It establishes the testing objectives, scope, approach, test environment, resources, schedule, deliverables, responsibilities, risks, traceability and acceptance criteria.

Scope: Testing covers the HMS residency lifecycle: admission and allocation, fees and payments, gate/outpass and attendance, mess, complaints, visitors, administration/reporting, security, performance, availability, backup/recovery, usability and accessibility. The SRS defines these functional and non-functional requirements and their acceptance tests.

References: Team 1 Software Requirements Specification (SRS), Team 1 Software Architecture Document (SAD), instructor Software Test Plan template, OWASP-based security validation, PCI-DSS payment-handling constraint, and WCAG 2.1 AA accessibility target.

Definitions: HMS — Hostel Management System; SRS — Software Requirements Specification; SAD — Software Architecture Document; RTM — Requirements Traceability Matrix; RBAC — Role-Based Access Control; MFA — Multi-Factor Authentication; UAT — User Acceptance Testing; SLA — Service Level Agreement; RPO/RTO — Recovery Point/Recovery Time Objectives.

2. Test Items

- Resident Portal — admission application, allocation status, fees/receipts, outpass, mess opt-out and complaints.
- Staff Back-Office — warden, security guard, maintenance staff, mess manager and administrator workflows.
- Gate Terminal Client — RFID/biometric entry-exit capture, manual fallback, local buffering and ordered replay.
- Auth & RBAC Service — authentication, authorization, session expiry and privileged MFA.
- Admission & Allocation Service — eligibility, bed allocation and transfer.
- Fee & Payment Service — invoice, payment, receipt and overdue processing.
- Gate & Outpass Service — movement records, outpass, curfew alerts and night attendance.
- Mess Service — weekly menu, opt-out and consumption-based billing.
- Complaint Service — lodging, assignment, lifecycle and SLA escalation.
- Visitor Service — visitor registration and visiting-hours enforcement.
- Reporting & Admin Service — reports, room inventory, no-dues and vacating/refund.
- Audit Service and PostgreSQL data layer — auditability, integrity and data retention.

3. Features to be Tested

Area	SRS IDs	Primary acceptance/test cases
Admission & Allocation	HMS-F-001 to F-003; F-061	TC-ADM-01, TC-ALL-01, TC-ALL-02, TC-INV-01
Fees & Payments	HMS-F-010 to F-013	TC-FEE-01 to TC-FEE-04

Gate / Outpass / Attendance	HMS-F-020 to F-023	TC-GATE-01, TC-GATE-02, TC-OUT-01, TC-ATT-01
Mess	HMS-F-030, F-031	TC-MESS-01, TC-MESS-02
Complaints	HMS-F-040 to F-042	TC-CMP-01 to TC-CMP-03
Visitors / Vacating	HMS-F-050, F-062	TC-VIS-01, TC-VAC-01
Reporting / Operations	HMS-F-060; NF-001 to NF-005	TC-RPT-01, TC-PERF-01, TC-OPS-01, TC-OPS-02
Security	HMS-SR-001 to SR-006	TC-SEC-01 to TC-SEC-06
Usability / Accessibility	HMS-NF-004	TC-UX-01

4. Features Not to be Tested

- Internal implementation of the institutional ERP; HMS only tests the documented read-only integration interface.
- Internal implementation of the payment gateway; HMS tests its integration, signed callback handling and the requirement that cardholder data is not stored.
- Internal implementation of the SMS/email provider; HMS tests notification requests, retry behaviour and failure handling at the interface.
- Internal firmware/design of the RFID or biometric reader; HMS tests the vendor event interface and gate-side handling.
- Third-party systems beyond the interfaces exposed to HMS, consistent with the SAD scope.

5. Test Approach / Strategy

5.1 Test Levels

- Unit testing — individual Django app modules, models, serializers, validation and business rules.
- Integration testing — REST APIs, PostgreSQL data access, ERP, payment, notification and gate-reader interfaces.
- System testing — end-to-end resident and staff workflows across the modular monolith.
- Acceptance testing / UAT — verification against the SRS acceptance criteria and representative hostel workflows.

5.2 Test Types

- Functional testing — verify all High/Medium/Low functional requirements.
- Regression testing — rerun impacted tests after fixes and changes.

- API testing — validate request/response contracts, authentication, authorization and error envelopes.
- Performance and load testing — verify the 2-second 95th-percentile target at 200 concurrent users with 2,000 seeded resident records.
- Security testing — TLS, RBAC, session expiry, MFA, audit logging, payment-data handling, input validation, SQL injection/XSS and OWASP-based vulnerability scanning.
- Usability and accessibility testing — mobile layout, keyboard navigation, WCAG 2.1 AA and task completion for low-skill users.
- Reliability/recovery testing — uptime evidence, nightly backup and restore drill.
- Compatibility testing — current Chrome, Edge, Firefox and Safari; desktop and Android/iOS mobile browsers.

Entry criteria: Stable build available; required modules deployed; test database and representative data available; test environment configured; external-system mocks/sandboxes or accessible interfaces available; test cases reviewed.

Exit criteria: 100% of planned test cases executed; every High-priority functional requirement implemented and verified; no non-functional or security requirement fails; all High-severity defects are closed; RTM shows Pass for every required test case.

5.5 Security Validation

- Verify TLS 1.2 or higher and reject weak/unsupported configurations.
- Test every role against restricted functions; unauthorized requests must return HTTP 403 where applicable.
- Verify 15-minute idle session expiry and token invalidation after logout.
- Verify Warden and Administrator MFA and salted password hashing using Argon2 (as specified by the SAD).
- Verify append-only/tamper-evident audit records for allocation, fee, outpass approval, override and account-management actions.
- Verify card numbers, CVV and bank credentials are not stored in HMS database or logs.
- Perform negative/fuzz input tests and OWASP-based checks for SQL injection and XSS.
- Verify sensitive resident, guardian, medical and identity-proof data is not exposed through unauthorized screens or API responses.

6. Test Environment

Category	Planned environment
Backend	Python 3.12, Django + Django REST Framework
Frontend	Django templates, Bootstrap 5, server-rendered responsive pages
Gate client	Server-rendered page + vanilla JS + local IndexedDB buffer
Database	PostgreSQL through Django ORM/migrations

External interfaces	ERP HTTPS/JSON sandbox or stub; PCI-DSS-compliant payment gateway sandbox/stub; SMS/email test service; RFID/biometric event stub/device SDK
Browsers/devices	Current Chrome, Edge, Firefox, Safari; desktop; Android/iOS mobile browsers; gate workstation
Performance	JMeter or equivalent load tool; production-like 200 concurrent users and 2,000 seeded records
API / UI tools	Postman/OpenAPI-style API tests; Selenium or equivalent UI automation where useful
Defect tracking	Jira
Reporting	PDF/XLSX generation validation using project reporting libraries

7. Test Schedule

The following schedule is proposed for the current academic project phase. Dates may be adjusted to match implementation progress.

Milestone	Planned date / period
Test plan preparation and review	16-Sep-2026
Test case design and RTM verification	17–18 Sep 2026
Test environment and test-data setup	18–19 Sep 2026
Unit and integration testing	20–23 Sep 2026
System / functional test execution	24–28 Sep 2026
Security, performance and accessibility testing	29 Sep–01 Oct 2026
Regression testing and defect verification	02–03 Oct 2026
UAT / acceptance review	04–05 Oct 2026
Test summary and sign-off	06 Oct 2026

8. Test Deliverables

- Software Test Plan (this document).

- Functional, API, security, performance and usability test cases.
- Automated/manual test scripts where applicable.
- Test data and seeded database.
- Test execution logs and screenshots/evidence.
- Defect reports and retest results.
- Requirements Traceability Matrix with test status.
- Final Test Summary Report and acceptance/sign-off record.

9. Roles and Responsibilities

Role	Name	Responsibility
QA Lead	Vaishnavi Goyal	Maintain the test plan, coordinate execution, monitor coverage and prepare the final summary.
Test Engineer	Yellala Sai Aneesh Reddy	Design and execute test cases, API/UI/security tests, capture evidence and log defects.
Developer	Vignesh Nunna	Provide builds, support environment setup, triage defects and implement fixes.
Product Owner	Vishnu S R	Validate acceptance criteria, review UAT results and support sign-off.

10. Risks and Mitigation

Risk	Impact	Mitigation
Unstable or delayed build	Test execution blocked	Request early smoke builds; prioritize critical-path tests first.
ERP/payment/notification dependency unavailable	Integration tests delayed	Use mocks/sandboxes and contract tests; run real integration tests when available.
Gate reader/device SDK unavailable	Gate tests blocked	Use a test event stub; later validate with the actual device interface.
Network outage at gate	Entry/exit events may be lost	Test IndexedDB buffering and ordered replay after reconnection.
Peak-season load	Response-time target may	Load test with 200 concurrent users and 2,000 seeded

	fail	records before acceptance.
Sensitive data exposure	Security/privacy impact	Use synthetic test data, verify RBAC/TLS/encryption/logging controls and run security tests.
Database failure	Data loss/outage	Verify nightly backups and quarterly restore procedure against RPO 24h/RTO 4h.

11. Assumptions & Dependencies

- The institutional ERP provides a stable read-only identity lookup interface or suitable test stub.
- The payment gateway provides a sandbox/test callback mechanism.
- SMS/email test delivery or a mock notification service is available.
- A gate-reader API stub/SDK is available when physical hardware is unavailable.
- Synthetic resident, guardian, fee, gate and complaint data will be used for testing.
- PostgreSQL test data can be reset between test runs.
- Required test tools and browsers are available to the team.
- The project will retain the architectural choices documented in the SAD unless formally revised.

12. Suspension & Resumption Criteria

Suspend testing if:

- Test environment is unavailable for more than 4 hours or cannot support the planned test run.
- The build is sufficiently unstable that more than 30% of planned tests are blocked.
- A critical defect causes data corruption, security compromise or prevents execution of major workflows.
- Required test dependencies are unavailable and no approved stub/workaround exists.

Resume testing if:

- Blocking/critical defects are resolved or an approved workaround is available.
- The environment is stabilized and test data is restored.
- Required external interfaces or test stubs are available.
- A smoke test confirms the build is suitable for continued execution.

13. Test Case Management & Traceability

The RTM in the SRS maps each requirement to a module and test case. The test plan uses the same requirement and test identifiers so that implementation, testing and acceptance remain traceable.

Requirement group	Test IDs / evidence
HMS-F-001 to F-003	TC-ADM-01; TC-ALL-01; TC-ALL-02
HMS-F-010 to F-013	TC-FEE-01 to TC-FEE-04
HMS-F-020 to F-023	TC-GATE-01; TC-GATE-02; TC-OUT-01; TC-ATT-01
HMS-F-030 to F-031	TC-MESS-01; TC-MESS-02
HMS-F-040 to F-042	TC-CMP-01 to TC-CMP-03
HMS-F-050, F-062	TC-VIS-01; TC-VAC-01
HMS-F-060 to F-061	TC-RPT-01; TC-INV-01
HMS-NF-001 to NF-005	TC-PERF-01; TC-OPS-01; TC-OPS-02; TC-UX-01; operations evidence
HMS-SR-001 to SR-006	TC-SEC-01 to TC-SEC-06

14. Test Metrics & Reporting

- Test case execution percentage.
- Pass/fail/blocked percentage.
- Requirement coverage — percentage of SRS requirements with executed tests and evidence.
- Defect density and severity distribution.
- Defect aging and retest status.
- Regression pass rate.
- Performance — 95th-percentile response time and concurrent-user capacity.
- Security — number and severity of findings from security validation.
- UAT completion and acceptance status.

Reports: Daily/iteration execution status, defect report, requirement coverage report, performance/security evidence, and Final Test Summary Report.

15. Representative Test Cases

Test ID	Scenario	Expected result
TC-ADM-01	Submit valid admission and then submit a duplicate for same enrolment/session.	Valid submission receives application number; duplicate is rejected with explanatory message.
TC-ALL-01	Allocate an approved applicant when vacant beds exist.	Only vacant beds matching gender, block and room type are considered and configured ranking is followed.
TC-ALL-02	Transfer resident to another bed.	Source becomes vacant, target occupied atomically; reason, actor and timestamp are audited.
TC-FEE-02	Process successful and failed/abandoned gateway callbacks.	Successful callback marks invoice Paid; failed/abandoned flow leaves invoice Unpaid.
TC-FEE-03	Generate receipt after successful payment.	Unique receipt number and downloadable PDF are available within 30 seconds.
TC-GATE-01	Scan resident entry/exit and test manual fallback.	Timestamped IN/OUT record is created for each movement.
TC-OUT-01	Resident requests outpass; warden approves/rejects.	State is Pending then Approved/Rejected; guardian notifications occur on approval and return.
TC-GATE-02	Resident is not recorded returned within curfew threshold.	Warden receives dashboard/push alert within 5 minutes of the 30-minute threshold.
TC-MESS-02	Attempt opt-out with less than 24 hours notice and calculate monthly bill.	Late opt-out is refused; bill is based on consumed days.
TC-CMP-02	Move complaint through lifecycle.	States follow Open ☐ In-progress ☐ Resolved ☐ Closed; closure requires complainant confirmation.
TC-CMP-03	Leave high/medium complaint unresolved beyond SLA.	Escalation is generated at 48h/96h respectively and logged.
TC-VIS-01	Register visitor inside and outside visiting hours.	Out-of-hours entry is refused without warden override and recorded reason; exit timestamp required.
TC-RPT-01	Generate management reports and export PDF/XLSX.	Reports reconcile with source records and both export formats open correctly.
TC-VAC-01	Vacate with incomplete and complete no-dues clearances.	Bed is released only after all required clearances; refund = deposit ☐ dues ☐ damages, negative result shown as payable.
TC-PERF-01	Load test resident portal/search with 200	95th percentile response time is ☐ 2 seconds.

	concurrent users and 2,000 records.	
TC-OPS-01	Perform database backup/restore drill.	Restore completes within 4 hours and data loss does not exceed 24 hours.
TC-OPS-02	Verify retention and anonymisation of old records.	Five-year-old records can be archived/retrieved as required and personal data is anonymised on schedule.
TC-UX-01	Untrained warden performs allocation; audit portal on mobile and keyboard.	80% of a 10-user trial completes allocation unaided within 5 minutes; WCAG 2.1 AA audit passes.
TC-SEC-01	Inspect TLS configuration.	No connection below TLS 1.2 and no weak ciphers.
TC-SEC-02	Test role restrictions, idle expiry and logout token reuse.	Unauthorized access returns 403; idle session and logged-out token are rejected.
TC-SEC-03	Inspect password storage and privileged login.	Passwords are non-reversible salted hashes; Warden/Admin MFA is mandatory.
TC-SEC-04	Perform privileged actions and inspect audit trail.	Actor, action, timestamp and outcome are recorded; entries cannot be edited through application.
TC-SEC-05	Inspect payment database/logs after payment flow.	No card number, CVV or bank credential is retained.
TC-SEC-06	Run injection/XSS security validation.	OWASP ZAP or equivalent reports no High/Medium injection or XSS findings.

16. Acceptance Test Suites

Suite	Coverage
TS-1 Admission and Allocation	TC-ADM-01, TC-ALL-01, TC-ALL-02, TC-INV-01
TS-2 Fees and Payments	TC-FEE-01 to TC-FEE-04
TS-3 Gate, Outpass and Attendance	TC-GATE-01, TC-GATE-02, TC-OUT-01, TC-ATT-01
TS-4 Mess	TC-MESS-01, TC-MESS-02
TS-5 Complaints	TC-CMP-01 to TC-CMP-03
TS-6 Visitors and Vacating	TC-VIS-01, TC-VAC-01
TS-7 Reporting and Operations	TC-RPT-01, TC-PERF-01, TC-OPS-01, TC-OPS-02
TS-8 Security	TC-SEC-01 to TC-SEC-06

17. Approvals

Role	Name	Signature / Date
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QA Lead

Development Lead

Product Owner

Document status: Draft for review. Final approval is subject to completion of implementation, test execution and the SRS exit criteria.